

Spoken Tutorial – Spoken Tutorial on Misinformation and Deepfakes

Title: Spoken Tutorial on Misinformation and Deepfakes

Module	AI Essentials
Episode	1. Misinformation and Deepfakes
Learning Objective	At the end of this tutorial learner will be able to 1. Define misinformation and deepfakes. 2. Explain how AI generates fake content. 3. Identify risks associated with deepfake technology. 4. Apply strategies for source verification and media literacy.
Approx. Duration	3-4 min
Outline	• How AI can generate realistic fake content • Risks of deepfake audio, video, and images • Real-world examples and societal impact • Importance of source verification and media literacy
Meta Tags	AI, deepfakes, misinformation, fake content, media literacy, source verification, societal impact, AI risks, synthetic media, digital ethics, AI generation, fake news
Pre-requisite Tutorial	Basic understanding of artificial intelligence concepts.

Script

Visual Cue	Narration
Title Slide	Welcome to this Spoken Tutorial on Misinformation and Deepfakes.
Learning Objectives Slide	In this tutorial, you will learn, • Define misinformation and deepfakes. • Explain how AI generates fake content. • Identify risks associated with deepfake technology. • Apply strategies for source verification and media literacy.
System Requirements Slide	Depends upon the requirement.
Pre-requisite Slide	To follow this tutorial, you should have a basic understanding of artificial intelligence concepts. No coding needed.
Section Header: Understanding AI-Generated Fake Content	Let's look at AI generated fake content. AI can create convincing fake content.
Illustration showing GANs with simple icons	What are GANs ? GANs are a primary technology. They generate fake media. They learn to create new data.

Illustration showing deep learning algorithms synthesizing audio, video, images	How does deep learning work? Algorithms are trained. They synthesize realistic audio. They also create video and images.
Illustration showing various types of AI-generated content like fake news, virtual influencers	What types of fake content exist? AI can produce fake news articles. It can create virtual influencers. Many forms of fake content are possible.
Section Header: The Dangers of Deepfakes	Now, let's move to the dangers of deepfakes. Deepfake technology has significant risks. It also has negative impacts.
Illustration showing broken public trust, media, and information sources	How do deepfakes affect trust? They erode public trust. Trust in media is lost. Trust in information sources is also lost.
Illustration showing manipulation of opinions, disinformation, and damaged reputations	Deepfakes can manipulate opinions. They spread disinformation. They can damage reputations. They affect individuals and narratives.
Illustration showing legal scales and ethical dilemmas related to deepfakes	What are the legal issues? Deepfakes pose legal challenges. They create ethical dilemmas. Proliferation of deepfakes is a concern.
Section Header: Deepfakes in the Real World	Let's look at deepfakes in the real world. We will explore actual instances. We will see societal consequences.
Illustration showing political figures, voting booths, and manipulated public opinion	How are deepfakes used politically? They influence political discourse. They impact elections. This can manipulate public opinion.
Illustration showing financial fraud, scams, and money deception	Deepfakes are used in fraud. They are employed in scams. They cause financial deception. Think about it.
Illustration showing a damaged reputation, discrediting public figures, and targeting private citizens	Deepfakes can damage reputations. They discredit public figures. They also target private citizens. This has serious consequences.
Screen close-up: user typing 'Ask for a misleading audio clip...'; AI generating response on screen	Open any AI tool. Type this in the message box. Ask for a misleading audio clip. Click Submit . Notice the quality. Is this clear or vague?
Screen close-up: user typing 'Ask for a convincing deepfake...'; AI generating response on screen	Let's improve it. Type for a convincing deepfake. Ask for realistic features. Click Submit . See how it could fool viewers. Compare this to the last example.
Side-by-side: real media vs deepfake output, highlighting subtle differences	Compare the outputs. We will visually compare real media. We will compare it with a deepfake. Notice the subtle tells. See the difference?

Section Header: Combating Misinformation: Your Role	Now, let's look at combating misinformation. We will discuss your role. You can identify deepfakes. You can counter misinformation.
Illustration showing icons for reverse image search, cross-referencing, and metadata check	How can you verify sources? Use reverse image search. Cross-reference information. Check metadata carefully.
Illustration showing a person with a thought bubble, questioning information, and evaluating evidence	Develop critical thinking skills. Question all information. Evaluate the evidence. Consider any biases.
Illustration showing icons for online tools, fact-checking websites, and browser extensions	What tools can help? Many online tools exist. Use fact-checking websites. Browser extensions are also useful.
Illustration showing media literacy promoting an informed society	Promote media literacy. It is a broader concept. It creates an informed society. This is very important.
Summary Slide	Let us summarize what we learned. We defined misinformation and deepfakes. We explored how AI generates fake content. We also discussed the risks. We learned strategies for verification and media literacy.
Assignment Slide	Now as an assignment, find two news articles. One real and one deepfake. Compare the results.
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